Sr. No. : AA / 1066933

SHRI RAMDEOBABA COLLEGE OF ENGINEERING & MANAGEMENT, NAGPUR.

(An Empowered Autonomous College of Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

This answer-book contains 16 pages. Additional answer-book will not be provided.

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Shri Ramdeobaba College of Engineering and Management, Nagpur

Test Examination I / II / III

Name of Student : Aniruddha P. Lanoki

Programme (Branch) : Data science Semester : 5th Shift :

Course (Subject) : ML Course Code : CDT 5002 Date of Examination 9/09/25

Roll No.							A	1	1	2	Student ID													

Signature of Invigilator :

Stamp of Signature of Exam. Incharge :

Examiner

Q. No.	Marks				
	a	b	c	d	Total
1	02				02
2	02	00			02
3	02	01			03
4					
5					
6					
				Marks Obtained	07

Grand Total in Words Seven

Name of Examiner : S.J. Signature

I have seen the answer book

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Q. No.	Marks				
	a	b	c	d	Total
1					
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3					
4					
5					
6					
				Marks Obtained	

Grand Total in Words

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Signature of Student

a)

→ 1) Decision Tree.

There are 6 instances inthe example where we will find Information Gain to decide main Node out of the two attributes.

let, training example = K.

Entropy of K =

$$K(+3,-3) = -\frac{3}{6}\log_2(3/6) - \frac{3}{6}\log_2(3/6)$$

$$= -0.5 \times -1 - 0.5 \times -1$$

$$= 0.5 + 0.5$$

$$= 1$$

∴ Entropy of training example = 1

Now Finding ~~entropy of~~ Information gain of attributes

$$IG = \text{Entropy} - \sum_{v=1}^{N} \frac{|S_v|}{|S|} (\text{entropy } v) \qquad (v = \text{attribute})$$

$$IG(K, a1) = 1 - \frac{3}{6} \times \left(-\frac{3}{6}(\log_2(3/6)) - \frac{3}{6}(\log_2(3/6))\right)$$

$$= 1 - \frac{3}{6} \times 1$$

$$= 1 - \frac{3}{6}$$

$$= 0.5$$

$$IG(K, a2) = 1 - \frac{4}{6} \times \left(-\frac{4}{6}(\log_2(4/6)) - 2/6(\log_2(2/6))\right)$$

$$= 1 - \frac{4}{6} \times 0.91$$

$$= 1 - 0.60$$

$$= 0.4$$

Since IG of a_1 is greater than a_2 a_1 will be the parent node

a_1

$\boxed{1, 2, 3}$ T $(+, +, -)$

$\boxed{4, 5, 6}$ F $(+, -, -)$ ... Can't decide

Now entropy of a_2

$a_2(+4, -2) = -\frac{4}{6}(\log_2(4/6)) - \frac{2}{6}(\log_2(2/6))$

$= 0.91$

$(a_2, T) = 1 - \frac{4}{6} \times \left(-\frac{4}{6}\log_2(4/6)\right)$

$= 0.91 - \frac{4}{6} \times 0.31 = 0.91 - 0.25 = 0.66$

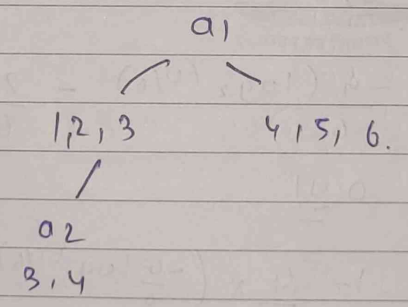
$(a_2, F) = 0.91 - \frac{2}{6}\left(-\frac{2}{6}\log^{(2/6)}\right)$

$= 0.91 - \frac{2}{6} \times 0.21$

$= 0.91 - 0.07$

$= 0.84$

Since IG of F is more next node will be F.



(1)

→ entropy of collection of training example with respect to the target function classification turns out to be 1.

(2)

→ Information gain of $a_1 = 0.5$
Information gain of $a_2 = 0.4$

Q2.
d]

→ In Regression we mainly focus on nearst value

Instance No.	X1		X2		Target Function (Y)	distance	Rank
1	1.2	2.77	2.3	1.44	1.8	2.05	7
2	3.4	0.81	1.5	4	3.2	2.19	8
3	2.2	0.09	3.1	0.16	2.9	0.5	1
4	1.9	0.36	2.8	0.09	2.3	0.92	4
5	3.1	0.36	4.0	0.25	3.5	0.78	3
6	2.0	0.25	3.5	0	2.7	0.5	2
7	1.5	1	1.0	0.25	1.4	1.11	5
8	3.7	1.44	3.2	0.09	3.8	1.23	6
9	2.5		3.5		?		

(1) Taking $k=3$

Instance 3, 6, 5 are 3 nearest Instance to Target function acc to regression would be mean of all target functions $= \frac{2.9+2.7+3.5}{3}$

$= \underline{\underline{3.03}}$

(2) Taking $K=5$

Instance 3, 6, 5, 4, 7 are 5 nearest

$$= \frac{2.9 + 2.7 + 3.5 + 2.3 + 1.4}{5} = \underline{\underline{2.56}}$$

0%

Algorithm:

(1)

Q2

b)

i) Increasing k mean we increase the no. of data set which will indeed tend the ans to more appropriate ans so. if the data would be under fitted so bais would be high if data is over fitted bais would be low & varien would be high so we do varience-bais trade of so lower varien but higher bais as more specific data is given.

~~True~~ False

ii) Since knn is a type of lazy learner we can see that the model learns during training so True !!

20/20

Q3.

d)

→ ① In perceptron we can only apply regression of linearly seperated ~~[illegible]~~ data set. But when the data set can't be linearly seperate we have to find some other way to over come this limitation.

② For applying regression model on non-linearly classified data we use gradient decent optimization algorithm.

③ Gradient descent optimization algorithm helps in reducing the error and optimizing the model which is continuously distributed.

④ Through partial differentiation of the change in Error weight we reach to the lowest error point in the data set.

(5) Derivation of gradient decent optimization algorithm

$$\Delta E = \frac{1}{2}(td - od)^2$$ — td is target ~~Error~~
od is actual ~~Error~~

$$\Delta w = -\eta\left(\frac{dE}{dw}\right)$$

$$\Delta wi = -\eta\left(\frac{dE}{dw}\right)$$

$$E(\Delta \bar{w}) = \left[\frac{dE_1}{dw_1}, \frac{dE_2}{dw_2}, \ldots \frac{dE_n}{dw_n}\right]$$

$$\frac{dE}{dw} = \frac{d}{dw} \frac{1}{2} \sum_{i=1}^{n} (td - od)^2$$

$$= \frac{1}{2} \sum \frac{d}{dw} (td - od)^2$$

$$= \frac{d}{dw}(td - od) - \frac{d}{dw}(td - od \times xi^0).$$

$$= \sum_{i=1}^{n} (td - od)(x_i)$$

$$\therefore \Delta w = n\,(td - od)\,x_i$$

n = optimization rate

td = target weight

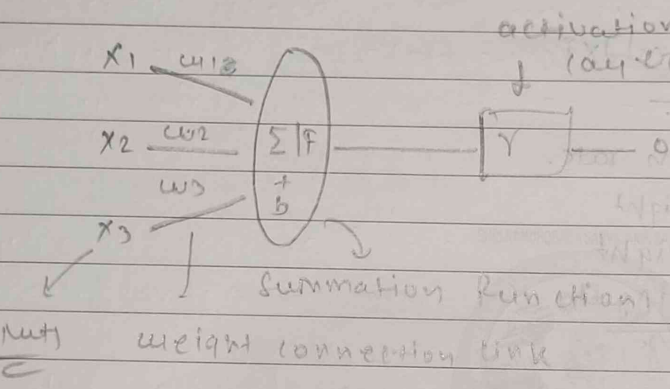
od = actual weight

x_i = current input

Q 2

Q3)

b)



(1)

in sigmoid activation function

$$y = \frac{1}{1+e^{-y}}$$

Inputs: $x_1 = 1.0$ $x_2 = -2.0$ $x_3 = 3.0$

$w_1 = 0.2$ $w_2 = -0.5$ $w_3 = 0.7$

$b = w_0 = 0.4$

$$y = (x_1 w_1 + x_2 w_2 + x_3 w_3) + b$$

$$= [0.1 \times 0.2 + (-2.0 \times -0.5) + (3.0 \times 0.7)] + 0.4$$

$= (0.02 + 1 + 2.1) + 0.4$

$y = 3.52$

$= \frac{1}{1+e^{-y}} = \frac{1}{1+e^{-3.52}} = \underline{\underline{0.971}}$

ii) tan h activation function.

$y = 3.52$

$h = 3.52$

$\tan(h) = \tan(3.52) = \underline{\underline{0.99999}}$